

Texas A&M University
College of Liberal Arts
Department of Economics
Intermediate Macroeconomics, ECON 410
Spring 2026, Section 503
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Midterm 1

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*On my honor, as an Aggie, I have neither given nor received unauthorized aid on
this academic work*

Signed: _____

NAME: _____

UIN: _____

37 questions in total. All questions have equal weight.
Please submit only your Scantron sheet and keep the question paper with you.

Consider a simple economy that produces and consumes two goods: food at home and cars. Due to the pandemic, consumers spend more time at home and eat at home rather than driving to a restaurant. Answer the following five questions based on the table below, which summarizes the prices and quantities in 2024 and 2025.

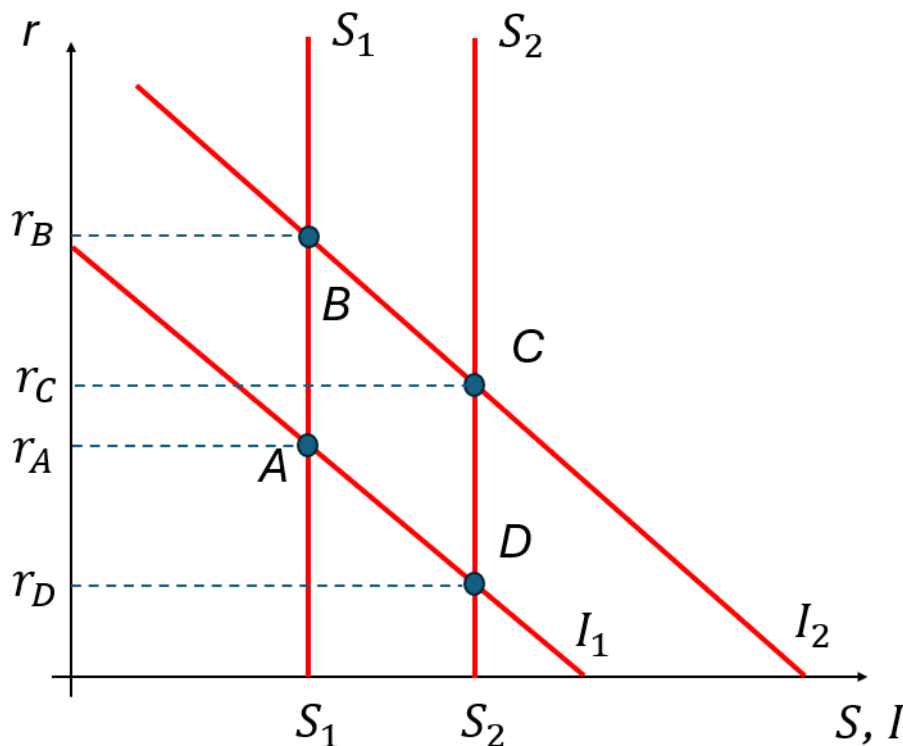
Category	2024		2025	
	Quantity	Price	Quantity	Price
Food at home	2	\$2	4	\$4
Cars	5	\$10	4	\$8

- Food at home inflation rate in 2025 is:
 - 50%
 - 0%
 - 20%
 - 100%
- Car price inflation rate in 2025 is:
 - 50%
 - 0%
 - 20%
 - 100%
- Set the base year as 2024. The real GDP in 2025 is:
 - 12
 - 24
 - 48
 - 68
- GDP deflator inflation rate (%) in 2025 is:
 - 1
 - 0
 - 48
 - 54
- Set base year as 2024, that means the consumption basket consists of 2 food at home and 5 cars. The CPI inflation rate (%) in 2025 is:
 - 11.1

- (b) 88.9
 - (c) -12.5
 - (d) 12.5
6. The Covid-19 Pandemic has led to lockdowns, mobility restrictions, and social-distancing rules that dramatically changed consumer expenditure patterns in many countries. In particular, consumers were spending less on transportation, hotels, restaurants, and recreation, while expenditures on food and other groceries have increased in both absolute and relative terms. Prices of the latter categories increased, while prices of the former categories decreased. These sudden changes in expenditure patterns made CPI overestimate inflation relative to the GDP deflator during the pandemic.
- (a) True
 - (b) False
7. During a typical recession, we observe unemployment
- (a) falling.
 - (b) rising.
 - (c) remaining constant.
 - (d) moving in either direction. There is no correlation between the two.
8. Assume a country has a population of 130 million adults where 100 million people are in the labor force and 30 million people are unemployed. Round to nearest whole number, the economy has an
- (a) unemployment rate equal to 30% and a labor force participation rate equal to 62%.
 - (b) unemployment rate equal to 23% and a labor force participation rate equal to 77%.
 - (c) unemployment rate equal to 30% and a labor force participation rate equal to 77%.
 - (d) unemployment rate equal to 23% and a labor force participation rate equal to 62%.
9. Based on the model we have learned from Chapter 3, if there is a decrease in taxes with no change in government spending, then public saving _____ and private saving _____.
- (a) increases; increases
 - (b) increases; decreases
 - (c) decreases; increases

- (d) decreases; does not change
10. Real GDP is a better measure of economic well-being than nominal GDP because real GDP:
- (a) excludes the value of goods and services exported abroad.
 - (b) includes the value of government transfer payments.
 - (c) measures changes in the quantity of goods and services produced by holding prices constant.
 - (d) adjusts the value of goods and services produced for changes in the foreign exchange rate.
11. An increase in the price of goods bought by firms and the government will show up in:
- (a) the CPI but not in the GDP deflator.
 - (b) the GDP deflator but not in the CPI.
 - (c) both the CPI and the GDP deflator.
 - (d) neither the CPI nor the GDP deflator.
12. Using a large-scale survey, economists study how individual transfers from the CARES Act affected consumption. In the two weeks following a sudden \$1,200 payment from the IRS, consumers immediately increased spending by an average of \$504, implying a marginal propensity to consume is _____. Ignoring equilibrium effects and assuming a constant MPC for each person, we estimate that the CARES Act's \$296 billion of payments to individuals will increase consumer spending by \$_____ billion. Round your solutions to second decimal points, for example, 3.5678 to 3.57.
- (a) 2.38, 704.48
 - (b) 0.42, 124.32
 - (c) 0.59, 174.64
 - (d) 2.38, 2856
13. In the simple model we studied in class, what brings in equilibrium the market of loanable funds and at the same time the market of goods and services?
- (a) Savings
 - (b) Real interest rate
 - (c) Consumption
 - (d) Taxes of the government

14. The economy begins in equilibrium at point A , representing the real interest rate r_A at which saving S_1 equals desired investment I_1 . What will be the new equilibrium combination of real interest rate, saving, and investment if the government cuts spending (G), holding other factors constant?



- (a) point A
 (b) point B
 (c) point C
 (d) point D
15. If output is described by the production function $Y = 5K + 7L$, then the production function has:
- (a) constant returns to scale.
 (b) diminishing returns to scale.
 (c) increasing returns to scale.
 (d) a degree of returns to scale that cannot be determined from the information given.
16. Suppose a firm uses one additional unit of capital and obtains 3 additional units of output. If the same firm instead uses two additional units of capital, it obtains 5 additional units of output. There is no change in the number of workers. Does the production function of the firm likely exhibit increasing, decreasing or constant marginal product of capital?

- (a) increasing.
 - (b) decreasing.
 - (c) constant.
17. When the government distributes stimulus paychecks, G _____, while T _____.
- (a) increases, stay constant.
 - (b) stay constant, increases.
 - (c) decreases, stay constant.
 - (d) stay constant, decreases.
18. Suppose that GDP (Y) is 5,000. Consumption (C) is given by the equation

$$C = 500 + 0.5(Y - T).$$

Investment (I) is given by the equation

$$I = 2,000 - 100r,$$

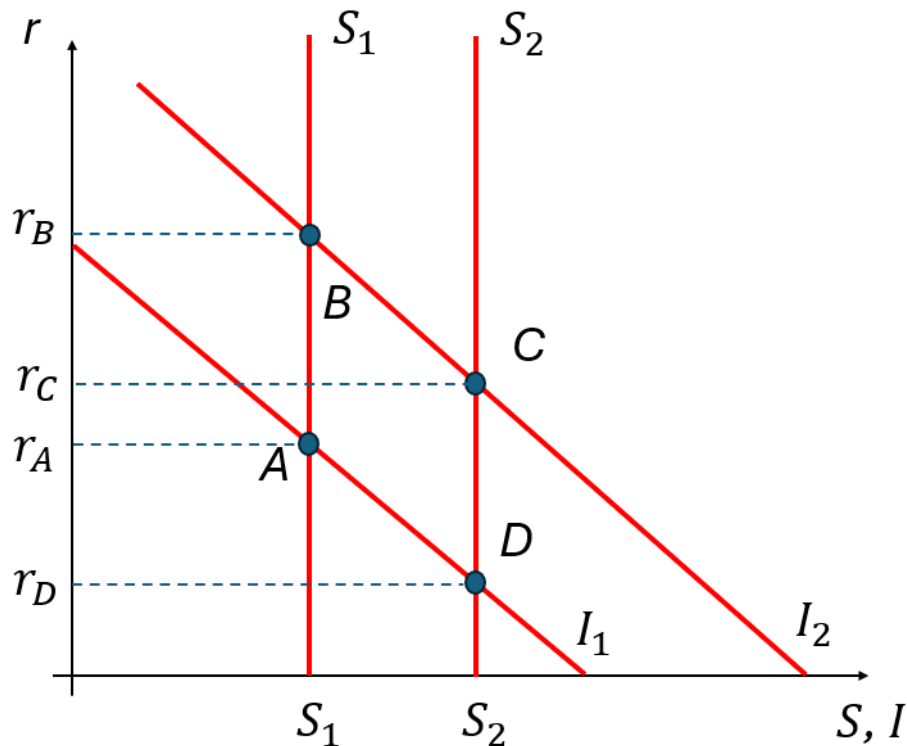
where r is the real interest rate, in percent. Government spending (G) is 1,000, and taxes (T) are also 1,000.

19. The equilibrium interest rate that equalizes saving and investment in the economy is:
- (a) 0%.
 - (b) 5%
 - (c) 25%
 - (d) 125%
20. Now, suppose we have the same setting as in the previous question, but additionally the government distributes stimulus paychecks worth 1,000. What would happen with I and r ? Assume that to do that, the government runs a bigger deficit (does not try to cut G).
- (a) Both I and r stay the same.
 - (b) I falls by 500 and r rises by 5 percentage points.
 - (c) I falls by 1000 and r rises by 10 percentage points.
 - (d) I falls by 1500 and r rises by 15 percentage points.
21. Deflation occurs when:
- (a) real GDP decreases.
 - (b) the unemployment rate decreases.

- (c) prices fall.
 - (d) prices increase, but at a slower rate.
22. In the actual economy real GDP _____ over time, and the growth rate of real GDP _____.
- (a) grows; fluctuates
 - (b) is steady; is steady
 - (c) grows; is steady
 - (d) is steady; fluctuates
23. If a person quits his job to become a stay-at-home parent, the labor-force participation rate _____ and the unemployment rate _____.
- (a) rises, rises
 - (b) rises, stays the same
 - (c) falls, rises
 - (d) falls, stays the same
24. Choose all transactions done in 2025 that raise consumption in 2025:
- A The Josh family purchases a new house that was built in 2025.
 - B Evan rents a one-bedroom apartment in College Station in 2025.
 - C A woman marries her butler in 2025. In 2024, before they were married, she paid him \$60,000 per year. He continues to wait on her as before (but as a husband rather than as a wage earner).
 - D Marcela purchases a 2008 100 trillion Zimbabwe dollar note from eBay.
- (a) A, B
 - (b) B, D
 - (c) A, B, D
 - (d) B
25. In an economy described by a Cobb–Douglas production function, workers should experience high rates of real wage growth when (*Hint: recall what real wage is equal to in equilibrium and think about how to interpret it*):
- (a) real interest rates are high.
 - (b) real interest rates are low.
 - (c) labor productivity is growing rapidly.
 - (d) capital's share of income is growing rapidly.

26. In a Cobb–Douglas production function, an increase in the capital share parameter α increases the marginal product of labor, holding capital and labor fixed and assuming $\frac{K}{L} < 1$.
- (a) True.
 - (b) False.
27. The government raises lump-sum taxes on income by \$100 billion, and output does not change. If the marginal propensity to consume is 0.6, private saving:
- (a) rises by \$40 billion.
 - (b) rises by \$60 billion.
 - (c) falls by \$60 billion.
 - (d) falls by \$40 billion.
28. The government raises lump-sum taxes on income by \$100 billion, and output does not change. If the marginal propensity to consume is 0.6, investment:
- (a) rises by \$100 billion.
 - (b) rises by \$60 billion.
 - (c) falls by \$60 billion.
 - (d) falls by \$100 billion.
29. When government spending increases and taxes are increased by an equal amount, interest rates:
- (a) increase.
 - (b) remain the same.
 - (c) decrease.
 - (d) can vary widely.
30. With a Cobb–Douglas production function, the share of output going to labor:
- (a) decreases as the amount of labor increases.
 - (b) increases as the amount of labor increases.
 - (c) increases as the amount of capital increases.
 - (d) is independent of the amount of labor.
31. Choose all people that an interviewer for the Current Population Survey would classify as out of labor force:

- A Jennifer Temple is working as a second-grade school teacher.
- B Frank Peabody is attending college full-time to earn a degree in elementary education.
- C Martin Hampton is working as a high school social science teacher but is at home sick with the flu.
- D Kyle Brown does not currently have a job. He wants to be an elementary-school teacher. He has the appropriate degree. He has not looked for a position in the past month because he doesn't believe schools are currently hiring.
- E Mark Powell was retired from Exxon Mobil.
- (a) B, C, D
- (b) B, C, E
- (c) B, C, D, E
- (d) B, D, E
32. A competitive, profit-maximizing firm hires labor until the:
- (a) marginal product of labor equals the wage.
- (b) price of output multiplied by the marginal product of labor equals the wage.
- (c) real wage equals the real rental price of capital.
- (d) wage equals the rental price of capital.
33. If $Y = AK^{0.5}L^{0.5}$ and A , K , and L are all 100, the marginal product of capital is:
- (a) 50.
- (b) 100.
- (c) 200.
- (d) 1000.
34. Who is included in the number of unemployed people in the US?
- (a) your neighbor, who stays at home and takes care of the children.
- (b) the other neighbor, who recently retired from teaching at the local high school.
- (c) a full-time student at TAMU.
- (d) none of the above.
35. The economy begins in equilibrium at point A , representing the real interest rate r_1 at which saving S_1 equals desired investment I_1 . What will be the new equilibrium combination of real interest rate, saving, and investment if there is a technological innovation that increases the demand for investment goods?



- (a) point A
 - (b) point B
 - (c) point C
 - (d) point D
36. Suppose an economy reports the following data for 2025 (in billions): Consumption: \$600, Investment: \$200, Firms profits: \$200, Government purchases: \$300, Government taxes: \$300 Exports: \$100, Imports: \$150. GDP in 2025 according to the expenditure approach is:
- (a) \$850
 - (b) \$750
 - (c) \$1550
 - (d) \$1050
37. Suppose GDP is \$100. The average wage is \$2 and there are 20 workers. Production uses only capital and labor, and factor markets are competitive. What is capital's share of income?
- (a) 20%
 - (b) 40%
 - (c) 60%
 - (d) 90%

